

IMS Design Report: Task 2

Team Name : Unpaid Interns

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1 Design Inspiration and Usability Principles

1.1 Design Inspiration: Influential Apps

The UI design of the Institute Management System (IMS) was inspired by three widely used applications, each contributing specific UI/UX patterns to our chosen modules (Dashboard, Timetable, and Attendance).

- **Google Calendar (Scheduling Experience):** The Timetable module draws heavily from Google Calendar. It utilizes a familiar horizontal date selector combined with a vertical time-block grid layout. This structure improves readability, allowing users to quickly understand daily classes and time slots without needing to learn a new scheduling format.
- **MyIIIT App (Dashboard Navigation & Utilities):** The Dashboard's utility grid and student view are inspired by the MyIIIT app. It employs a clean, icon-based modular grid (3×3) for primary navigation. Paired with the required central search bar, it ensures that critical features are visually accessible immediately upon login.
- **IRCTC App (Spatial Selection & Filtering):** The Attendance module uses a spatial selection metaphor inspired by the IRCTC app's train coach and seat selection interface. By presenting the class roster as an interactive visual grid, instructors can filter and toggle student states (Present/Absent) much like selecting available seats, making bulk marking highly efficient.

1.2 Application of Usability Principles

Based on the foundational UI/UX guidelines, the prototype implements the following core usability principles:

- **Short Term Memory (STM) Optimization:** Human short-term memory is limited to roughly 7 ± 2 items. To prevent cognitive overload, the application organizes the Dashboard into a hierarchical utility grid. Users are not forced to memorize complex, deeply nested menu paths to find the Attendance or Timetable modules.
- **Consistency and Standards:** The interface adheres to established industry conventions to ensure a shallow learning curve. By mirroring Google Calendar's temporal grid for the Timetable, the system leverages the user's existing mental models. Users instinctively know how to read and interact with the schedule because it behaves exactly like the tools they already use.
- **Error Prevention:** The system proactively minimizes input mistakes. Dense lists of student names can lead to accidental mis-clicks during roll call. By utilizing the IRCTC-inspired attendance grid, the UI replaces traditional lists with distinct, color-coded visual blocks. This spatial separation and clear state visualization drastically reduce the probability of marking the wrong student absent.

2 Wireframes and Mockups

In accordance with the project requirements, both Low-Fidelity wireframes and High-Fidelity Figma mockups have been designed for the Dashboard, Timetable, and Attendance modules. These prototypes define the complete end-to-end user flows and visual hierarchy, and are provided in the attached design submission.

Figma Design File: The complete interactive prototype, low-fidelity wireframes, and high-fidelity mockups can be accessed here:
<https://www.figma.com/design/pEgAW5q4MTdloWEAJOlLet/IMS-App?node-id=0-1t=X6FwzbuZwvo8kPq6-1>